

SLOPEWATCH · REQUIREMENTS

SlopeWatch requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	SLW-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/slopedwatch		
Source	docs/03-requirements.md @ e2af6fe		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

SlopeWatch requirements

These are first-pass requirements for one monitored site. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see SLW-PRB-001). The status column gives the position against the first-order estimates in SLW-PRC-001; "met" means met on paper only.

The **reference site** is a slope or dump face up to about 50 m wide with three tilt stakes on the fall line, one crack gauge across a tension crack at the crest, and a mast with the FieldNode core and alert unit at the toe, within 60 m of cable.

Table 1. SlopeWatch requirements for one site.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measure surface tilt	Resolution 0.01 degrees or finer; range ± 30 degrees or more on two axes	Sensor datasheet; later bench tilt table	Met (datasheet)
R2	Limit false tilt from temperature	Apparent tilt change from the daily soil temperature cycle 0.02 degrees or less per day, and 0.002 degrees per hour or less, without software correction	Thermal estimate; later chamber test (CalRig)	Met, thin margin (estimate)
R3	Measure crack opening	Range 100 mm or more; resolution 0.1 mm; re-settable in the field without tools beyond a spanner	Datasheet and design review	Met (datasheet class)
R4	Sample and report often enough to see acceleration	Every sensor read every 10 min or faster; uplink every 60 min in the normal state and every 10 min in the precaution or warning state	Firmware sketch review; airtime estimate	Met
R5	Local alarm without a network	Siren and beacon start within 60 s of a warning condition, driven by the on-site node alone	Design review; later bench test	Met by design
R6	Remote alert	At least two named people notified by SMS or app within 5 min of a warning, where gateway and mobile coverage exist	Design review; later end-to-end test	Met only where coverage exists; not met at sites without a gateway
R7	Alarm audible where people work	65 dB(A) or more at 100 m from the mast in open ground	Spreading-loss estimate; later field measurement	At risk: about 65 to 70 dB(A) estimated in quiet ground; not met near running machinery
R8	Energy autonomy	5 days with no sun, including one 30 min alarm, on the FieldNode cell	Energy estimate	Met on energy; at risk on FieldNode 12 V rail current (open question)
R9	Survive burial and weather	Capsule and stake head IP67 and buried to 0.4 m; node IP65; operate -10 to 50 °C; cable in conduit rated for burial and UV	Datasheets and design review	Unverified
R10	Installable by a small team	Each stake installed by two trained people with hand tools (post-hole auger or driven pilot, hand-mixed grout) in 45 min or less; no work below an actively moving face	Method review; later timed trial	Unverified
R11	Affordable	Parts for one reference site \$250 or less	Priced BOM (bom/ bom.csv)	Not met: about \$380 with the FieldNode core; about \$254 without it
R12	Open, local data	90 days or more of raw readings kept on the node; open CSV export; works with any LoRaWAN server	Design review	Met (FieldNode flash)
R13	Trustworthy alarms		Field trial with partner	

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
		No more than one false warning per site per year, and a documented statement of the failure modes the system cannot detect		Not verifiable at TRL 2; statement drafted in SLW-PRC-001

■ Assumptions

- Tilt thresholds of 0.01 degrees per hour (precaution) and 0.1 degrees per hour (warning) follow Uchimura et al. (2015); site-specific thresholds must be set with a geotechnical engineer after a baseline period.
- The FieldNode core provides about 19 Wh of LiFePO₄ storage (3.2 V, 6 Ah), a switched 5 V and 12 V sensor rail, SPI flash and a LoRaWAN radio, as described in the FieldNode README.
- Siren sound level 110 dB(A) at 1 m, from typical 12 V piezo sirens; free-field spreading of 6 dB per doubling of distance.
- Daily temperature range at the soil surface about 20 °C, damped to about 10 to 15 % of that at 0.3 m depth in moist soil (estimate).
- Stake spacing and site layout will change after co-design; R11 is stated for the reference site only.